



Data Engineer

Interview – NTT DATA

North America

(Mixed Scenario Set)



1. What they asked me:

Design a pipeline to ingest data from multiple sources (API + DB + files) into a data warehouse.

What I said:

I would design a layered architecture:

- Source → Landing → Raw → Curated → Warehouse
- Use API connectors + CDC for DB + batch file ingestion
- Store raw data in object storage (S3/ADLS)
- Use Spark or ADF/Glue for transformation
- Apply schema validation and data quality checks
- Load curated data into warehouse (Snowflake/Redshift/Synapse)

I also mentioned retry logic, logging, and alerting.

Tips:

Always talk about **layers, scalability, monitoring, and failure handling**.

2. What they asked me:

Your Spark job is running very slow — what will you check first?

What I said:

I check:

- Data skew
- Shuffle partitions
- Join strategy
- Stage DAG in Spark UI
- Memory & executor config
- Broadcast join opportunities
- File sizes (small file problem)

Then I tune partitions and join type.

Tips:

Mention **Spark UI + skew + shuffle tuning** — interviewers expect this.

3. What they asked me:

How do you implement incremental loads?

What I said:

Using:

- Timestamp column
- Auto-increment ID
- CDC logs
- Watermark tracking table

I store last processed value and load only new/updated records.

Tips:

Always mention **watermark table** — very strong answer.

4. What they asked me:

How do you ensure data quality in pipelines?

What I said:

I implement:

- Schema validation
- Null checks
- Range checks
- Duplicate detection
- Referential integrity checks
- Row count reconciliation

Bad records go to quarantine tables.

Tips:

Say **reject + quarantine + audit table** — very practical.

6. What they asked me:

A query with joins is taking 20 minutes — how do you optimize?

What I said:

I check:

- Execution plan
- Index usage
- Join type
- Filter pushdown
- Partition pruning
- Remove SELECT *
- Materialized views if needed

Tips:

Always say **EXPLAIN PLAN** — must-have keyword.

6. What they asked me:

Explain SCD Type 2 with example.

What I said:

SCD Type 2 keeps history by:

- Adding new row when change happens
- Using effective_date and end_date
- Marking active flag

Example: customer address change → new record created.

Tips:

Use **real business example** — customers/products.

7. What they asked me:

What if late records arrive after aggregation is done?

What I said:

I reprocess affected partitions or use:

- Delta merge
- Upserts
- Window reprocessing
- Backfill jobs

Tips:

Mention **idempotent pipeline design**.

8. What they asked me:

How do you choose partition columns?

What I said:

Based on:

- Query patterns
- High cardinality vs low
- Filter columns
- Date-based partition for large fact tables

Tips:

Say **query-driven partitioning** — strong answer.

9. What they asked me:

When would you choose streaming over batch?

What I said:

Streaming when:

- Near real-time needed
- Fraud detection
- IoT events
- Monitoring systems

Batch when:

- Daily reporting
- Historical analytics

Tips:

Use **use-case driven answer**.

10. What they asked me:

How do you handle skewed keys?

What I said:

I use:

- Salting keys
- Broadcast joins
- Repartition
- Skew hints (Spark)

Tips:

Salting example = interviewer favorite.

11. What they asked me:

Pipeline failed mid-run — how do you recover?

What I said:

- Check checkpoint/state
- Restart from last successful stage
- Use idempotent writes
- Use job bookmarking

Tips:

Say **restartable + idempotent pipeline**.

12. What they asked me:

How do you organize folders in data lake?

What I said:

I use:

```
/raw/source/date  
/processed/domain/table  
/curated/business_layer
```

Plus versioning and schema folders.

Tips:

Mention **medallion architecture (bronze/silver/gold)**.

13. What they asked me:

Explain one pipeline you built end-to-end.

What I said:

I explained:

- Source → ingestion → transform → validation → load
- Tools used
- Data volume
- Performance tuning
- Monitoring approach
- Business impact

Tips:

Use numbers: **data size, runtime, improvement %**.

14. What they asked me:

Star vs Snowflake schema?

What I said:

Star = denormalized → faster queries

Snowflake = normalized → less redundancy

I prefer star for analytics.

Tips:

Keep it practical, not theoretical.

15. What they asked me:

How do you make pipelines production ready?

What I said:

- Logging
- Monitoring
- Alerting
- Retry logic
- SLA tracking
- Data validation
- Documentation

Tips:

Production readiness = big scoring answer.